Supplementary material

Study	Selection		Comparability	Outcome
	Random Sequence Generation	Allocation Concealment		
Young 2018*	Low risk	Low risk	Low risk	Low risk

Study	Selection	Comparability	Outcome	Total
Ageno 2017	* * * *		* * *	7
Nicklaus 2017	* * * *	* *	* *	8
Ageno 2017	* * * *	* *	* *	8
Chaudhury 2017	* * * *	* *	* *	8
Boiles 2019	* * * *	* *	* *	8
Simmons 2018	* * * *	* *	* *	8
Hummert 2017	* * *	* *	* *	7
Signorelli 2017	* * * *	* *	* *	8
Khorana 2018	* * *	* *	* *	7
Angelini 2018	* * *	* *	* *	7
Streiff 2018	* * *	* *	* *	7
Lee JH 2020	* * * *	*	* * *	8
Lee JH 2020	* * * *	*	* * *	8
Lee JH 2019	* * * *	*	* * *	8
Lee JH 2019	* * * *	*	* * *	8
Wysokinski 2019	* * * *	* *	* *	8

Table 1: Risk of bias assessment of included studies

*Young 2018 is a randomized controlled trial (RCT). It was evaluated using the Cochrane tool for RCT. All other studies are observational cohorts and were evaluated utilizing the New-Castle Ottawa (NCS) quality assessment tool.

Table 2: Sensitivity analysis of studies evaluating the net clinical benefit.

Excluded study	Pooled RR	LCI 95%	HCI 95%	Cochran Q	p	I 2	I 2 LCI 95%	I 2 HCI 95%
Young 2018	0.817589	0.74686	0.895015	10.49915	0.724855	0	0	38.13651
Ageno 2017	0.819871	0.74941	0.896958	9.962094	0.76495	0	0	34.80143
Lee JH-1 2019	0.81152	0.741913	0.887657	7.503154	0.913574	0	0	13.43449
Lee JH-2 2019	0.818556	0.748153	0.895584	10.36524	0.735022	0	0	37.33727
Lee JH-3 2020	0.818695	0.748471	0.895509	10.20134	0.74732	0	0	36.3305
Lee JH-4 2020	0.813613	0.74369	0.890109	9.79501	0.777012	0	0	33.68926
Nicklaus 2018	0.818575	0.748351	0.89539	10.25358	0.743419	0	0	36.65489
Ross 2017	0.81546	0.745412	0.89209	10.35573	0.735741	0	0	37.27971
Simmons 2018	0.817182	0.747018	0.893935	10.5066	0.724287	0	0	38.18034
Signorelli 2017	0.815246	0.745434	0.891596	9.941886	0.76642	0	0	34.6689
Hummert 2017	0.815208	0.745195	0.891799	10.29007	0.740683	0	0	36.87953
Chaudhury 2018	0.817658	0.747438	0.894475	10.47285	0.72686	0	0	37.98115
Wysokinski 2019	0.814225	0.743864	0.891241	10.27824	0.741571	0	0	36.80687
Boiles 2019	0.82163	0.751205	0.898657	7.705071	0.904118	0	0	15.703
Khorana 2019	0.876407	0.75138	1.022238	9.303575	0.811126	0	0	30.18658
Streiff 2018	0.79609	0.721025	0.878969	9.102673	0.824418	0	0	28.64576

Table 3: Sensitivity analysis of studies evaluating the VTE recurrence.

Excluded study	Pooled OR	LCI 95%	HCI 95%	Cochran Q	p	I^2	I^2 LCI 95%	I^2 HCI 95%
Young 2018	0.785709	0.706253	0.874104	11.38803	0.65532	0	0	42.96516
Ageno 2017	0.788708	0.709474	0.876791	10.91617	0.692609	0	0	40.49979
Lee JH-1 2019	0.778308	0.700313	0.864988	8.156629	0.881007	0	0	20.36976
Lee JH-2 2019	0.787072	0.707978	0.875003	11.26734	0.664918	0	0	42.35424
Lee JH-3 2020	0.788446	0.709543	0.876124	10.39773	0.732565	0	0	37.53309
Lee JH-4 2020	0.781289	0.702872	0.868454	10.63428	0.714497	0	0	38.92258
Nicklaus 2018	0.787132	0.708285	0.874756	11.14105	0.674922	0	0	41.70079
Ross 2017	0.783649	0.705074	0.870981	11.23808	0.66724	0	0	42.20418
Simmons 2018	0.785366	0.706568	0.872952	11.39135	0.655056	0	0	42.98178
Signorelli 2017	0.783328	0.705009	0.870348	10.81315	0.700648	0	0	39.93293
Hummert 2017	0.783024	0.70446	0.87035	11.13438	0.675449	0	0	41.66586
Chaudhury 2018	0.786027	0.707183	0.873663	11.36208	0.657386	0	0	42.83493
Wysokinski 2019	0.781555	0.702577	0.869411	11.09531	0.678533	0	0	41.46048
Boiles 2019	0.790967	0.711801	0.878937	8.341402	0.870792	0	0	22.13367
Khorana 2019	0.847915	0.705727	1.018752	10.3923	0.732976	0	0	37.50045
Streiff 2018	0.765921	0.682827	0.859126	10.26893	0.742269	0	0	36.74956

Table 4: Sensitivity analysis of studies evaluating the MB events.

Excluded study	Pooled RR	LCI 95%	HCI 95%	Cochran Q	p	I 2	I 2 LCI 95%	I 2 HCI 95%
Young 2018	1.03337	0.829084	1.287993	15.45506	0.347772	9.414753	0	46.84597
Ageno 2017	1.078932	0.87086	1.336719	15.3896	0.352051	9.029479	0	46.3101
Lee JH-1 2019	1.049856	0.837868	1.315479	15.88574	0.320402	11.87066	0	49.77483
Lee JH-2 2019	1.095492	0.871847	1.376506	15.84408	0.322989	11.63893	0	49.52774
Lee JH-3 2020	1.086343	0.853073	1.383399	16.74429	0.270061	16.38942	0	53.83041
Lee JH-4 2020	1.060294	0.8354	1.345731	16.48095	0.284893	15.05345	0	52.7525
Nicklaus 2018	1.081816	0.853801	1.370723	16.6025	0.27798	15.67536	0	53.26422
Ross 2017	1.072146	0.836823	1.373645	16.85866	0.263788	16.95662	0	54.26552
Simmons 2018	1.062296	0.834381	1.352468	16.58729	0.278839	15.59801	0	53.20156
Signorelli 2017	1.055909	0.838951	1.328975	16.14166	0.3048	13.26789	0	51.16751
Hummert 2017	1.073622	0.83722	1.376777	16.88139	0.262554	17.06842	0	54.34987
Chaudhury 2018	1.052342	0.840075	1.318243	15.91606	0.318527	12.03854	0	49.95074
Wysokinski 2019	1.090218	0.843939	1.408367	16.9209	0.260417	17.2621	0	54.49495
Boiles 2019	1.023896	0.8699	1.205153	13.88222	0.458522	0	0	53.21252
Khorana 2019	1.220044	0.93707	1.588471	13.70616	0.471824	0	0	52.61151
Streiff 2018	0.940204	0.790655	1.118039	12.0235	0.604416	0	0	45.97958

Table 5: Sensitivity analysis of studies evaluating the CRNMB events.

Excluded study	Pooled RR	LCI 95%	HCI 95%	Cochran Q	p	I 2	I 2 LCI 95%	I 2 HCI 95%
Young et al. 2018	1.841545	1.337474	2.535592	7.62728	0.366604	8.224164	0	70.24449
Lee JH-1 2019	2.121561	1.455644	3.092117	9.583284	0.213443	26.95615	0	67.08203
Lee JH-2 2019	2.183806	1.480004	3.222294	8.950947	0.256197	21.79599	0	63.80276
Lee JH-3 2020	2.12969	1.487083	3.049983	9.261758	0.234405	24.4204	0	65.55418
Lee JH-4 2020	2.093958	1.442982	3.038612	9.76756	0.202133	28.3342	0	67.85899
Ross 2017	2.0987	1.503335	2.929847	8.839046	0.264419	20.80593	0	63.08538
Simmons 2018	1.927765	1.431694	2.595719	7.518882	0.37692	6.901051	0	69.81551
Chaudhury 2018	2.035224	1.3999	2.95888	9.911608	0.193638	29.37574	0	68.42527
Wysokinski 2019	1.83113	1.365531	2.455481	6.93064	0.436137	0	0	67.25359

Table 6: Sensitivity analysis of studies evaluating the mortality data.

Excluded study	Pooled RR	LCI 95%	HCI 95%	Cochran Q	p	I 2	I 2 LCI 95%	I 2 HCI 95%
Young et al. 2018	0.694308	0.530904	0.908006	32.87441	1.11E-05	81.74872	63.404	90.89766
Ageno 2017	0.809466	0.67357	0.97278	17.906	0.006471	66.49168	25.17867	84.99348
Lee JH-1 2019	0.660898	0.483247	0.903857	30.10233	3.76E-05	80.06799	59.39535	90.21577
Lee JH-2 2019	0.771334	0.611353	0.97318	23.57212	0.000626	74.54621	45.81484	88.04293
Lee JH-3 2020	0.678118	0.510188	0.901324	32.69847	1.20E-05	81.65052	63.1716	90.8575
Lee JH-4 2020	0.662237	0.504966	0.868489	28.63318	7.14E-05	79.04529	56.92514	89.80612
Simmons 2018	0.736288	0.580439	0.933984	31.53646	2.00E-05	80.9744	61.56536	90.58211
Wysokinski 2019	0.745685	0.583257	0.953348	29.64622	4.59E-05	79.76133	58.65703	90.09254

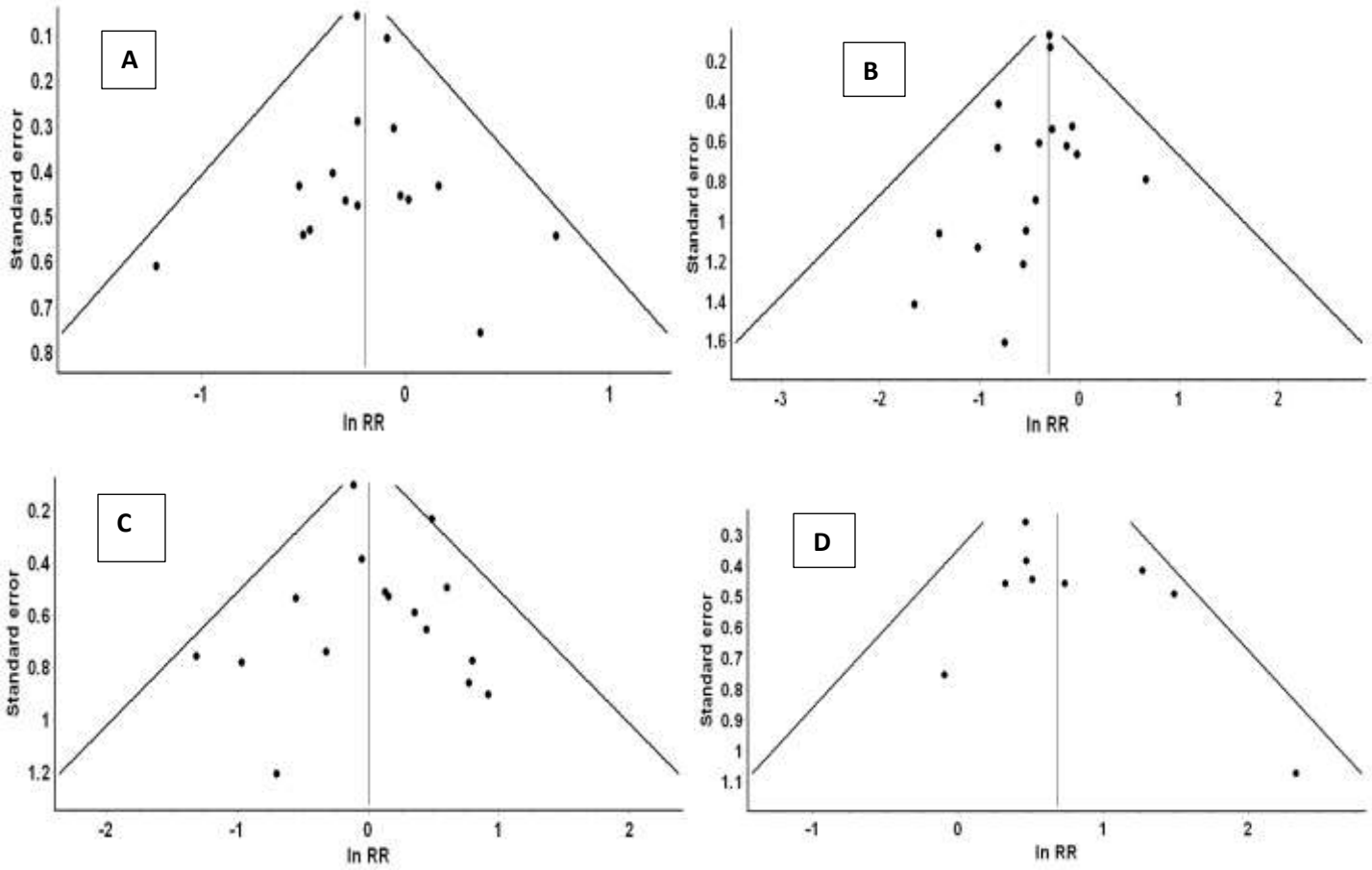
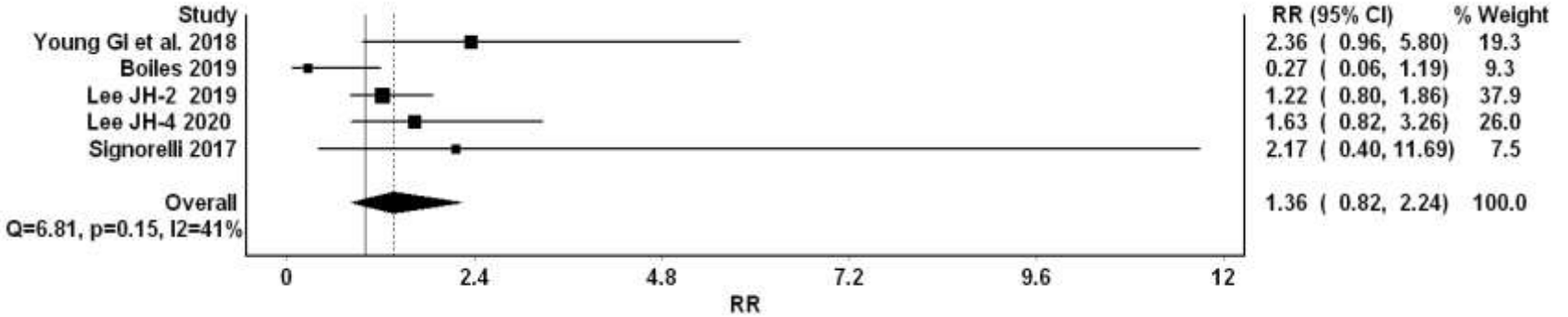


Figure 1: Funnel plots of studies evaluating various outcomes in rivaroxaban users vs. LMWH.

#**(A)** NCB (showing no asymmetry), **(B)** VTE recurrence (Marked asymmetry), **(C)** MB events (no asymmetry), and **(D)** CRNMB (mild asymmetry). The absence of asymmetry suggests less likelihood of publication bias.

A



B

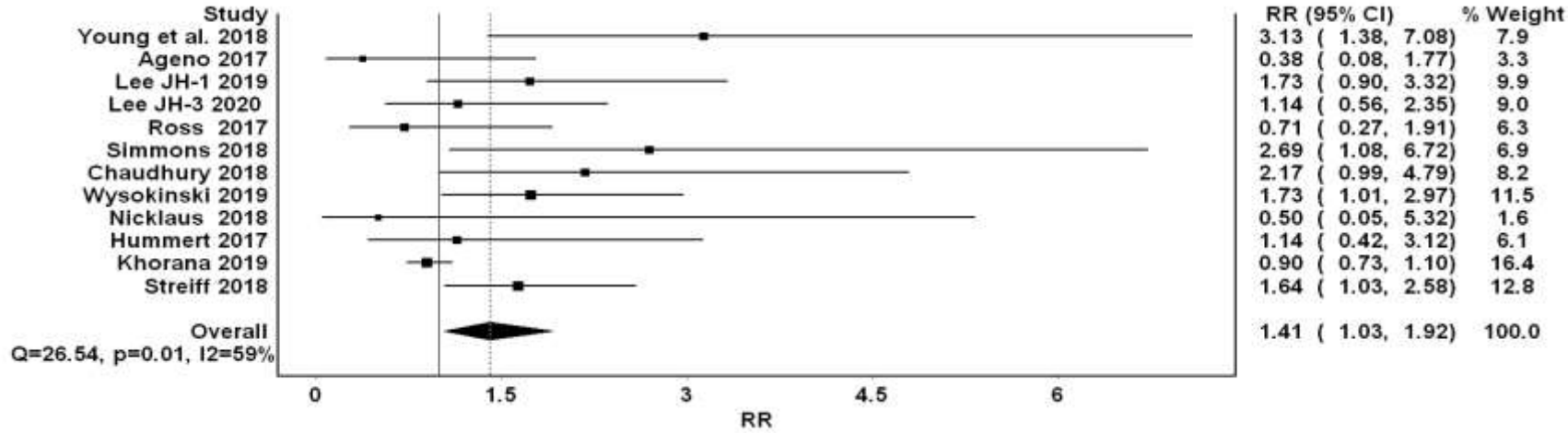
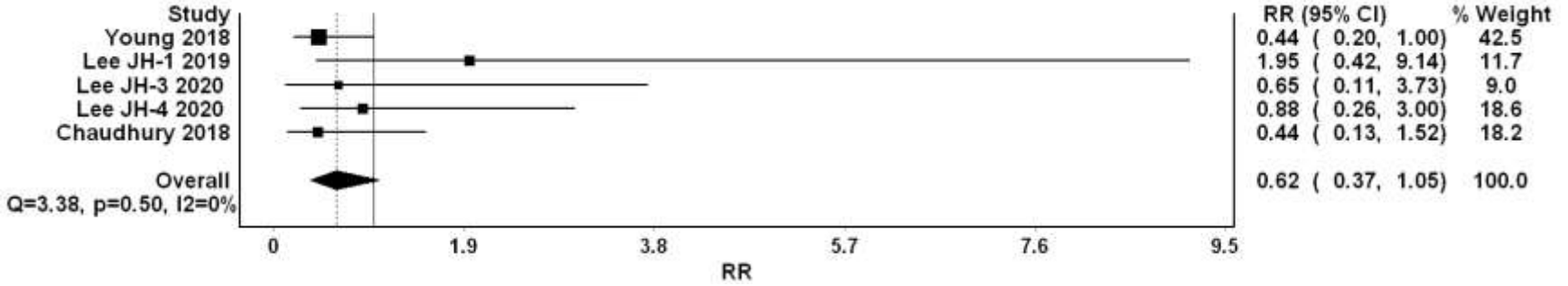


Figure 2: Forest plot comparing bleeding events (MB or composite of MB & CRNMB whenever reported) in rivaroxaban and LMWH users among studies (A) restricted to GI or GU malignancies, and (B) studies non-restricted to GI or GU malignancies. There was a significantly increased risk of composite bleeds in non-restricted studies (B).

MB=Major bleeding, CRNMB= clinically relevant non major bleeding, GI= gastrointestinal, GU=genitourinary.

A



B

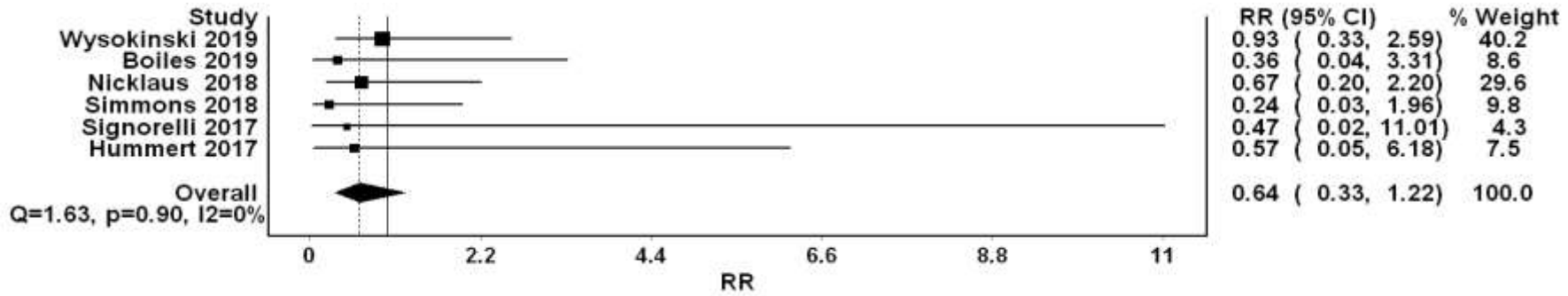
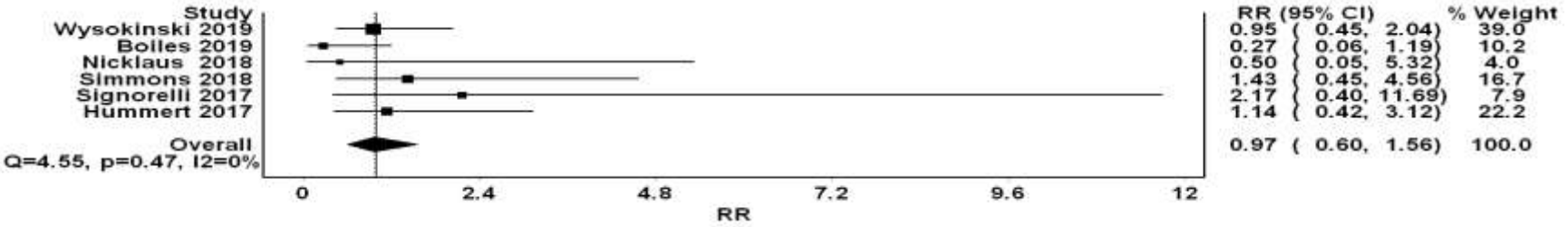


Figure 3: Forest plot comparing VTE recurrence in rivaroxaban user's vs. (A) dalteparin, and (B) enoxaparin. There is a protective trend (likely underpowered) with rivaroxaban compared with both LMWH formulations. This trend was more apparent compared with dalteparin. The results are homogenous evident by I^2 of 0%.

A



B



C

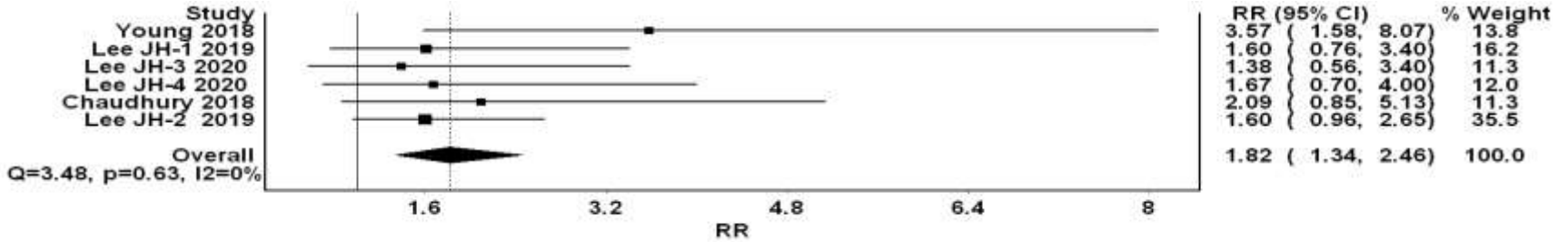


Figure 4: Forest plot comparing MB events in rivaroxaban user's vs. (A) dalteparin, and (B) enoxaparin. Additionally, (C) CRNMB among rivaroxaban and dalteparin users. There is a trend towards rivaroxaban having more MB events compared with dalteparin, while there was no evidence of a difference between rivaroxaban and enoxaparin in terms of MB. CRNMB events were higher in the rivaroxaban group compared to dalteparin. The results are homogenous evident by I^2 of 0%. There were no enough studies to compare CRNMB in the rivaroxaban group and enoxaparin.

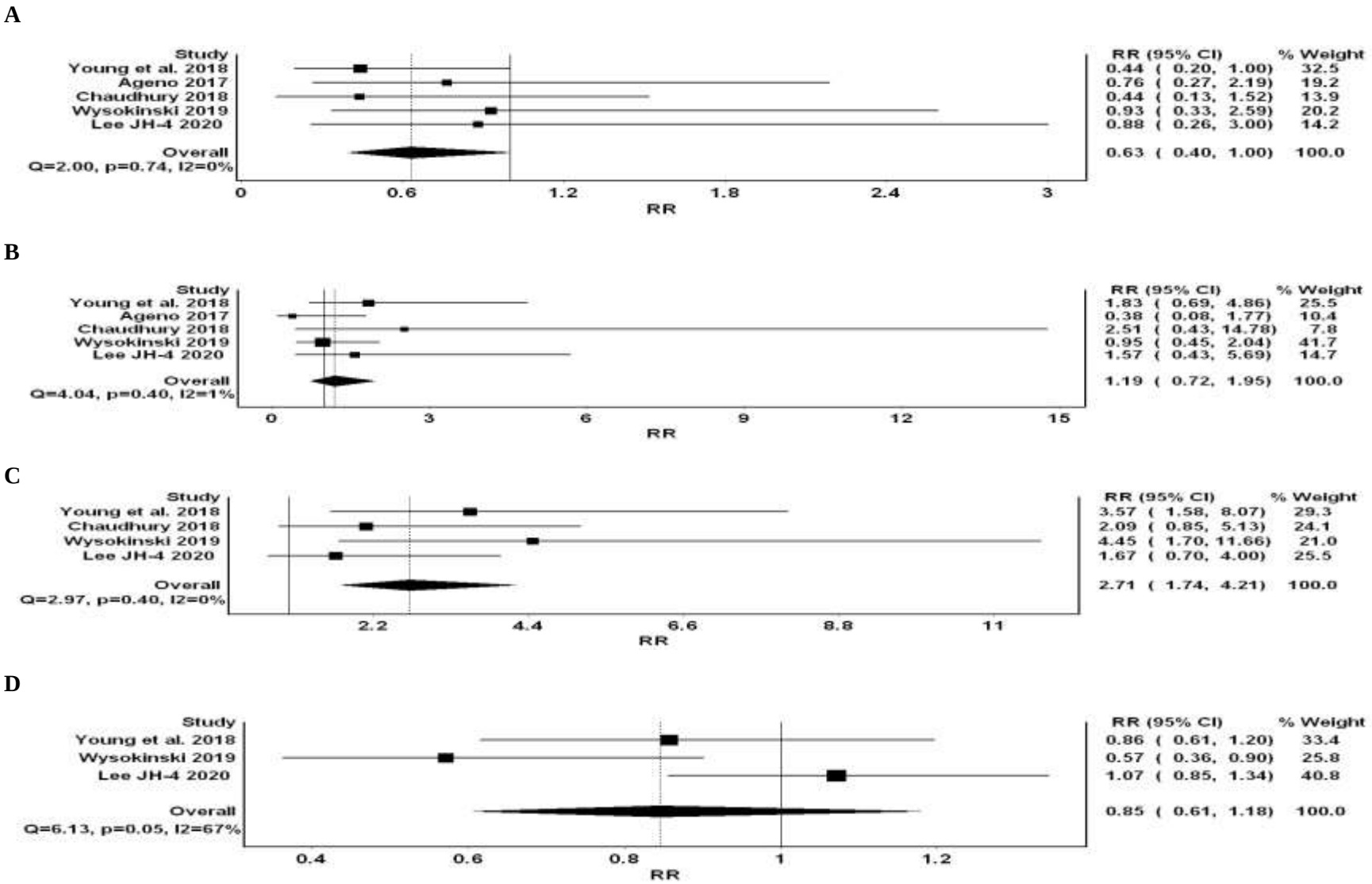


Figure 5: Forest plot comparing (A) VTE recurrence, (B) MB, (C) CRNMB, and (D) all-cause mortality events in rivaroxaban users compared with LMWH utilizing studies with equal therapeutic duration. The mortality rate was not different between both groups with marked heterogeneity, while the remaining outcomes' results were consistent with the principal analysis.